

TECHWIZ 7 CHAMPIONSHIP

EGREEN BASKET THEME



MarketLink

Farm Fresh Just a Click Away

A unified full-stack web application connecting certified local farmers market growers directly with neighborhood consumers. Features predictable weekly recurring harvest stock, 0% gateway in-person pickup settlement, interactive OpenStreetMap navigation, and 24/7 AI Intelligent Assistance.

Project Name:	MarketLink	Version:	1.0 (Final Official Release)
Championship:	Aptech TechWiz 7	Category:	End-to-End Web Solutions
Technology Stack:	React 18 • Bootstrap 5 & Custom CSS (No Tailwind) • Node.js • Express • MongoDB		
Interactive Elements:	ReactBits (Custom Cursor, DecryptedText, BentoGrid,		

SpotlightCard) •
Three.js 3D
Basket •
OpenStreetMap

Documentation Date:

September
2026

Deliverables:

PDF,
Word
(.docx &
.doc),
Plain
Text
(.txt),
SQL
Script

1. Executive Summary & Problem Definition

1.1 Background & Necessity

Local farmers markets are growing rapidly across urban and suburban communities as health-conscious shoppers seek fresh, seasonal, and locally grown agricultural produce. Despite rising popularity, traditional market operations suffer from acute communication bottlenecks and logistical friction:

- **Shopper Uncertainty & Wasted Trips:** Consumers rarely know in advance which farmers will attend a particular market day, what varieties they will bring, or at what prices. Popular seasonal items (such as organic berries or sourdough) frequently sell out within the first 30 minutes, resulting in wasted commutes and frustration.
- **Farmer Harvest Miscalculation & Waste:** Growers lack pre-harvest demand visibility. Over-harvesting causes unsold perishable surplus at the end of hot market days, causing agricultural food waste. Under-harvesting deprives farmers of potential revenue.
- **Lack of Spatial Coordination:** Stalls are informally distributed, making it challenging for customers with tight schedules to locate vendors and pick up orders quickly.

💡 **The MarketLink Vision:** MarketLink transforms local farmers markets by establishing a digital pre-order loop. Farmers publish weekly recurring inventory templates prior to harvest day, and customers reserve produce baskets with specific morning pickup windows. Pre-orders are paid for in-person at pickup, protecting growers from credit card gateway deductions and ensuring fair trade.

1.2 Core Objectives

100%

GUARANTEED
PRODUCE PICKUP

0%

ONLINE GATEWAY
DEDUCTION

30+

FARMERS SUPPORTED

24/7

AI MARKET
ASSISTANCE

1.3 Out-of-Scope Clarifications (As per SRS Page 8)

In strict compliance with the TechWiz 7 SRS guidelines:

- **No Online Payment Gateways:** Payment for pre-orders is settled 100% in person at market pickup (Cash, Card, or Mobile Pay at the stall).
- **No Courier / Delivery Logistics:** The platform explicitly supports market-day pickup only, eliminating delivery carbon emissions.
- **No Food Safety / Certification Auditing:** Government licensing verification is handled by physical market organizers outside the application scope.

2. System Architecture & UI/UX Design System

2.1 Multi-Tier Architecture

MarketLink implements a resilient, clean multi-tier web architecture designed for responsiveness, high performance, and rapid maintainability:

Layer	Technology Stack	Architectural Function
Presentation (Frontend)	React 18, Vite, Bootstrap 5, Custom CSS	Single Page Application (SPA), reactive state management, custom cursor follower, responsive layouts across mobile/desktop. Strictly Zero Tailwind CSS.
Interactive UI (ReactBits)	ReactBits Components, GSAP, CSS Animations	DecryptedText cyber-organic headings, SpotlightCard mouse-following radial glows, BentoGrid ecosystem cards, MagnetButton interactions, animated counters.
3D Graphics Engine	Three.js Canvas Engine	Interactive 3D harvest basket containing organic fruits, vegetables, ambient golden pollen particles, and cursor-tracking tilt responsiveness.
Mapping & Geolocation	Leaflet, OpenStreetMap API	Interactive community market locator with custom markers, popup details, coordinates, and turn-by-turn routing directions.
Application Server (Backend)	Node.js, Express.js REST API	Stateless REST endpoints, JWT authorization, stock auto-deduction and restoration, AI assistant integration, notification dispatch.
Database & Persistence	MongoDB Native + Mongoose ODM	Scalable document storage with automated schema validation and embedded persistent memory fallback for instant zero-config evaluation.

2.2 UI/UX Philosophy: The "eGreen Basket" Aesthetic

The visual interface is crafted around a cohesive organic nature theme:

- Color Palette:** Deep Forest Green (#064e3b), Fresh Emerald (#10b981), Mint Light (#ecfdf5), Golden Amber (#f59e0b), Warm Earth (#78350f).
- Typography:** *Outfit* for modern, approachable editorial headings and *Plus Jakarta Sans* for crisp legibility across body text and tables.
- Glassmorphism:** Translucent glass cards with `backdrop-filter: blur(14px)` and soft green ambient glows.

3. Functional Requirements & Role Breakdown

3.1 Customer Features

- **Registration & Authentication:** Full profile with name, email, phone, and address. Role-based login with JWT security.
- **Browse Markets & Interactive Map:** Filter markets by day (Saturday, Sunday, Wednesday) and city. View stall locations on OpenStreetMap with direct navigation links.
- **Produce Catalog & Multi-Criteria Filtering:** Filter by Category (Vegetables, Fruits, Dairy, Bakery, Honey, Herbs), price range, market, and in-stock availability.
- **Pre-Order for Market Pickup:** Add produce to harvest basket, choose pickup date and morning time window (e.g. 09:00 AM – 10:00 AM). Stock is automatically reserved.
- **Order Management & Modification:** Track status timeline (Placed → Accepted → Ready for Pickup → Completed). Modify pickup time slot or cancel before farmer cutoff time.
- **1-Click Re-Order:** Instant re-order feature for recurring weekly staples.
- **Favorites & Restock Alerts:** Bookmark favorite farmers and staple produce for quick weekly access.
- **Ratings & Reviews:** Post 1–5 star reviews with comments after order completion.

3.2 Farmer / Vendor Features

- **Stall Profile Management:** Set stall name, contact person, market location, operating days, pickup time windows, and cutoff hours.
- **Weekly Stock & Pricing Management:** Full CRUD operations for products (name, category, price, unit, stock quantity, harvest day, photo). Support for recurring weekly templates.
- **Toggle Sold-Out Status:** 1-click toggle to mark items sold out or restore availability as harvest updates.
- **Pre-Order Fulfillment Workflow:** View incoming customer pre-orders, accept or decline with cancellation reasons, mark orders "Ready for Pickup", and record in-person completion.
- **Farmer Sales Insights:** Real-time KPI summaries: Total Revenue, Pending Orders, Active Listings, and Best-Selling items.
- **Customer Review Responder:** View customer reviews and post public grower responses.

3.3 Administrator Features

- **Executive KPI Dashboard:** Platform metrics: Total Farmers, Pending Approvals, Total Customers, Active Markets, Total Pre-Orders, and Revenue across all hubs.
- **Farmer Verification & Approval:** View new farmer registrations and approve or suspend them before they can list products.
- **Customer Account Moderation:** Activate or suspend customer accounts in case of policy violations.
- **Farmers Markets CRUD:** Add, edit, or remove community markets with GPS coordinates, operating schedules, and directions.
- **Content Moderation:** Inspect and remove inappropriate product listings or customer reviews.

- **Global Announcements:** Broadcast platform-wide alerts and market news to farmers and customers.

3.4 Built-in AI Market Assistant (24/7)

An intelligent, domain-trained chatbot embedded in every page. It queries live database records to answer customer questions about:

- Operating hours and days for each farmers market.
- Farmer attendance, stall numbers, and pickup time windows.
- Product availability (e.g., "Find organic heirloom tomatoes").
- How pre-orders work and payment settlement at the stall.

4. Database Schema & Test Credentials

4.1 Relational Schema & MongoDB Document Mapping

Entity / Collection	Primary Key / Identifier	Foreign Relationships	Key Attributes
Users	<code>_id / user_id</code>	—	name, username, email, password_hash, role ('admin','farmer','customer'), status, stallName, pickupTimeWindows.
Markets	<code>_id / market_id</code>	—	name, address, city, operatingDays, timings, latitude, longitude, map_provider, stallCount, image.
Products	<code>_id / product_id</code>	<code>farmerId → Users</code>	name, category, price, unit, stock_quantity, image, isSoldOut, isRecurringTemplate, harvestDay, ratingAverage.
Orders	<code>_id / order_id</code>	<code>customerId,</code> <code>farmerId → Users</code>	orderNumber, items [product, qty, price], total_amount, order_status, pickupDate, pickupTimeSlot, paymentStatus.
Reviews	<code>_id / review_id</code>	<code>productId,</code> <code>farmerId,</code> <code>customerId</code>	rating (1–5), comment, farmerReply, farmerReplyAt, createdAt.
Announcements	<code>_id / announcement_id</code>	—	title, message, audience ('all','farmers','customers'), type, active.

4.2 User Credentials Matrix (MANDATORY TechWiz Requirement)

 **Evaluator Quick Login:** The application header includes a "  Quick Demo Roles" dropdown allowing evaluators to switch between Admin, Farmer, and Customer with 1 click without typing passwords.

Role	Full Name	Email / Username	Password	Access Scope & Permissions
Admin	Platform Administrator	<code>admin@marketlink.com</code> username: admin	<code>admin123</code>	Full governance, farmer approvals, customer activation, markets CRUD, moderation, sales reports.

Role	Full Name	Email / Username	Password	Access Scope & Permissions
Farmer 1	Ramesh Kumar (Green Valley)	greenvalley@marketlink.com username: farmer_john	farmer123	Weekly recurring stock management, pre-order fulfillment, mark ready, review replies.
Farmer 2	Lakshmi Devi (Sunny Meadow)	sunnyorchard@marketlink.com	farmer123	Orchard fruit & berry stock management, morning pickup window settings.
Pending Farmer	New Organic Grower	newfarmer@marketlink.com	farmer123	Pending verification account (Used to test the Admin 1-Click Approval workflow).
Customer 1	Priya Sharma	priya@marketlink.com username: priya_s	customer123	Browse markets, map directions, place pre-orders, choose pickup slots, modify/cancel, reviews.
Customer 2	Rahul Verma	rahul@marketlink.com	customer123	Fresh harvest pre-orders, saved favorite farms, and order history.

5. Project Installation & Execution Instructions

5.1 System Prerequisites

- **Node.js:** v18.0.0 or higher (Tested on Node.js v24.18.0)
- **Database:** MongoDB (Local daemon or MongoDB Atlas). If MongoDB is offline, the backend's automatic persistent JSON fallback engine activates seamlessly.
- **Browser:** Chrome, Edge, or Firefox.

5.2 Step-by-Step Launch Commands

Step 1: Start the Backend REST API Server

```
cd "d:\MarketLink\Source Code\backend"
node server.js
```

Output: 🚀 MarketLink Server running on: http://localhost:5000

Step 2: Start the React Frontend Application

```
cd "d:\MarketLink\Source Code\frontend"
npm run dev
```

Output: → Local: http://localhost:3000/

5.3 Evaluation Live Links

Target Component	URL	Description
Frontend Web Application	http://localhost:3000	React 18 SPA with 3D Basket, Leaflet Map, ReactBits components.
Backend REST API	http://localhost:5000	Express API with MongoDB native connectivity.
System Health Check	http://localhost:5000/api/health	Reports server health, theme name, and active database mode.

5.4 Compliance Checklist

- ☒ **Zero Tailwind CSS:** Crafted with pure Bootstrap 5 and handcrafted custom CSS.
- ☒ **React & MongoDB:** React 18 client talking to Node/Express REST API backed by MongoDB.

-  **ReactBits Interactive UI:** Custom cursor, DecryptedText, BentoGrid, SpotlightCard, MagnetButton.
-  **Three.js 3D Animations:** Interactive 3D harvest basket with ambient particles and cursor tracking.
-  **OpenStreetMap Integration:** Live coordinates, stall markers, and route navigation.
-  **In-Person Settlement:** Pre-orders settled at pickup in compliance with SRS scope.
-  **Multi-Format Documentation:** PDF, Word (.docx & .doc), Plain Text (.txt), and SQL DDL scripts.